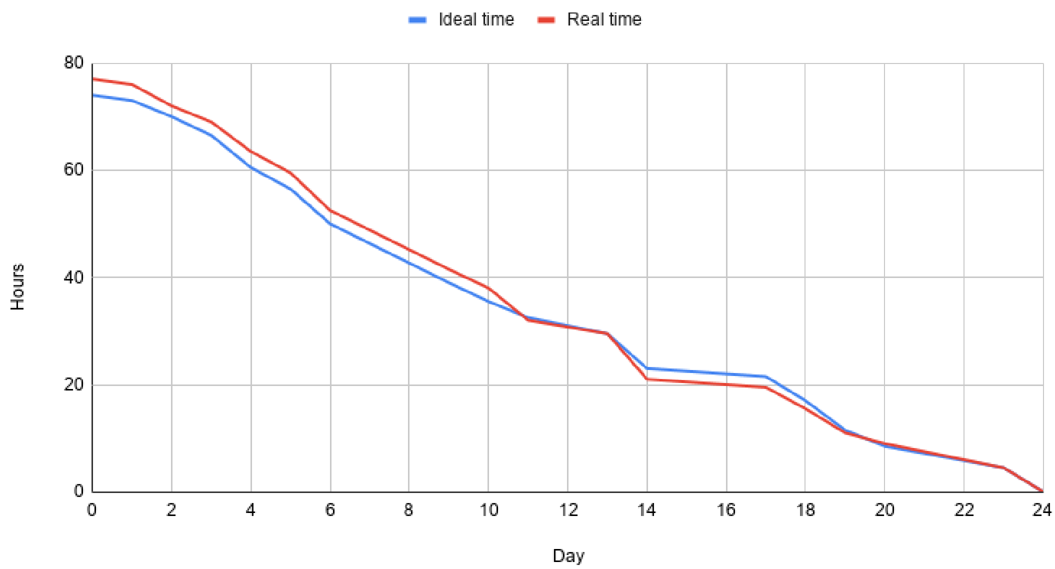
Sprint 3 - Review

Implemented features/completed work + design decisions (narrative)

- front-end implementation
- Backend updates
- Data infrastructure
- Deployment
- Optional features (login, AI chatbot, routing)

Burndown chart



This sprint started slightly above the ideal pace, reflecting the higher volume of work planned for this extended sprint compared to the previous sprints. Progress tracked closely to the ideal time for the first 12 days, from which point progress was completed at a faster pace than we had originally estimated. All tasks were completed on schedule but by the end had taken several hours more to complete than the ideal time predicted.

Sprint review

Sprint 3 was the longest and most feature-intensive sprint of the project, as it extended over a three week period instead of the two weeks for the other sprints. Work was divided into four main sections, covering both frontend and backend development, data infrastructure and upload, and the implementation of the optional features (user login, AI chatbot and route planning options).

In terms of the frontend, the scaffolding and routing were set up, reusable UI components were created and the data visualisation pages were implemented alongside filtering and search controls. The frontend was integrated with the backend API, with the QA and UX polishes carried out towards the end of the sprint. Backend tasks focused on extending the data access layer, developing additional API endpoints for visualisations and improving the performance of the application. Unit and integration testing of the backend was also carried out during this sprint. Another core part of this sprint was preparing, cleaning and uploading the historical data to EC2 for use in the machine learning model in the next sprint. This involved choosing a storage approach, provisioning the environment and verifying the uploaded data.

It was decided that we would implement all three of the optional features during this sprint. The authentication system, AI chatbot and route planning were implemented in the frontend, which allowed for the option to include all features should we have the time to do so.

Sprint retrospective

Overall, this sprint went well and we made strong progress across frontend, backend, and deployment tasks. We divided responsibilities clearly at the start, which helped ensure all parts of the system were covered. The team worked well together and most tasks were completed on time, including API development, frontend integration, authentication, and deployment to EC2. A key strength of this sprint was how we handled more complex and connected tasks compared to the previous sprint. Features like integrating the frontend with backend APIs and deploying both components required coordination between team members, which was managed effectively. We also made good progress on major features such as data visualisation, authentication, and AI integration. One area for improvement is task tracking and clarity around ownership as the workload became more complex. While tasks were assigned, there were some overlaps and occasional uncertainty around progress, especially where tasks depended on each other. This sometimes led to small delays or the need for clarification during meetings. For the next sprint, we aim to improve visibility by updating the backlog more consistently and sharing progress in real time through our group chat. This should reduce confusion and help coordination, particularly for dependent tasks. Overall, the team maintained a positive dynamic and successfully completed a wide range of tasks this sprint.